8e. Barrier Functions

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Code Script

This section's scripts are available **here**, under the name `allCode.jl`. They've been tested under Julia 1.11.8.

INTRODUCTION

This section presents an approach to mitigating type instability through the use of barrier functions. A **barrier function** is a type-stable function that's called from within a type-unstable function, with the variables of uncertain type passed explicitly as arguments. Introducing barrier functions prompts the compiler to infer concrete types for those variables, effectively creating a "barrier" that prevents the propagation of type instability to subsequent operations.

A key benefit of barrier functions is that **they're agnostic to the underlying source of type instability**, making them widely applicable across scenarios.

> ⚠️ **Warning!** - Barrier Functions Should Be Considered as a Second Option
>
> Barrier functions are preferred for situations where type instability is either difficult to fix or directly unavoidable. Keep in mind that the original function will remain type unstable, entailing different consequences depending on the instability nature. For this reason, it's best to aim for type-stable code from the outset whenever possible.

APPLYING BARRIER FUNCTIONS

To illustrate the technique, let's revisit a type-unstable function from a previous section. This defines a variable `y` based on `x`, subsequently performing an operation involving `y`.

▼ Type Instability

TYPE INSTABILITY

```julia
function foo(x)
    y = (x < 0) ?  0  :  x

    [y * i for i in 1:100]
end

@code_warntype foo(1)        # type stable
@code_warntype foo(1.0)      # type UNSTABLE
```

In the example, `0` is an `Int64`, whereas `x` could be either an `Int64` or `Float64`. This leads to two possibilities:

- if `x` is an `Int64`, then `y` will also be an `Int64`, making `foo(1)` type stable.
- if `x` is a `Float64`, the compiler then can't determine whether `y` will be an `Int64` or a `Float64`, rendering `foo(1.0)` type unstable.

A barrier function can address the type instability of the second case. It requires embedding a type-stable function into `foo`, passing `y` as an argument. The function will then attempt to deduce `y`'s type, allowing the compiler to use this information for subsequent operations. The example below defines `operation` as a barrier function. [1]

Barrier Function

BARRIER FUNCTION

```julia
operation(y) = [y * i for i in 1:100]

function foo(x)
    y = (x < 0) ?  0  :  x

    operation(y)
end

@code_warntype operation(1)     # barrier function is type stable
@code_warntype operation(1.0)   # barrier function is type stable

@code_warntype foo(1)           # type stable
@code_warntype foo(1.0)         # barrier-function solution
```

W `Float64`. Nevertheless, this ambiguity no longer matters, as `operation(y)` will determine the type of `y` before the array comprehension is executed. As a result, the expression `[y * i for i in 1:100]` will be computed with a method specialized for the specific type of `y`, ensuring type stability.

> ⚠ **Warning!**
>
> Barrier Functions should address the type instability *before* the type-unstable operation is executed. Otherwise, we're back to the original issue, where the compiler has to check `y`'s type at each iteration and select a method accordingly.
>
> For example, `foo` in the example below doesn't apply the technique correctly: `y` can be either `Float64` or `Int64`, but `operation(y,i)` only identifies the type inside the for-loop. Thus, the compiler is forced to check `y`'s type at each iteration, which is the original problem we intended to solve.

Wrong Use

WRONG USE

```
operation(y,i) = y * i

function foo(x)
    y = (x < 0) ?  0  :  x

    [operation(y,i) for i in 1:100]
end

@code_warntype foo(1)          # type stable
@code_warntype foo(1.0)        # type UNSTABLE
```

REMARKS ON @CODE_WARNTYPE

Functions that introduce barrier functions are capable of addressing type instability. However, this effect isn't necessarily reflected when `@code_warntype` is executed. The reason is that barrier functions typically mitigate type instability, rather than completely eliminating it. When this is the case, nonetheless, the impact of the remaining type instability may be negligible. And even if the barrier function successfully eliminates the type instability, a red warning may still be triggered.

To illustrate this, let's start presenting a scenario where the barrier function completely eliminates the type instability, yet a red warning shows up.

Type Unstable | Barrier Function

TYPE UNSTABLE

```
x = ["a", 1]                    # variable with type 'Any'



function foo(x)
    y = x[2]

    [y * i for i in 1:100]
end
```

```
julia> @code_warntype foo(x)
```

```
MethodInstance for foo(::Vector{Any})
  from foo(x)
Arguments
  #self#::Core.Const(Main.foo)
  x::Vector{Any}
Locals
  #62::var"#62#63"
  y::Any
Body::Vector
1 ─       (y = Base.getindex(x, 2))
│   %2  = Main.:(var"#62#63")::Core.Const(var"#62#63")
│   %3  = y::Any
│   %4  = Core.typeof(%3)::DataType
│   %5  = Core.apply_type(%2, %4)::Type{var"#62#63"{_A}} where _A
│   %6  = y::Any
│         (#62 = %new(%5, %6))
│   %8  = #62::var"#62#63"
│   %9  = (1:100)::Core.Const(1:100)
│   %10 = Base.Generator(%8, %9)::Base.Generator{UnitRange{Int64}, F} where F<:var"#62#63"
│   %11 = Base.collect(%10)::Vector
└──       return %11
```

In this example, `y` is defined from an object with type `Vector{Any}`. This leads to a red warning, as `x[2]` has type `Any` and therefore the compiler can't infer a concrete type for `y`. However, no operation is involved at that point, as we're only performing an assignment. Since the only operation performed uses a barrier function, the lack of type information is inconsequential. Overall, type instability is never impacting performance after introducing a barrier function.

In contrast, the example below demonstrates that a barrier function may only alleviate type instability, rather than eliminate it entirely. In this scenario, the operation `2 * x[2]` is type unstable, forcing the compiler to generate code for each possible concrete type of `x[2]`. Nonetheless, this operation has a negligible performance impact on `foo`, justifying why the barrier function only targets the more demanding operation.

Type Unstable | Barrier Function | Type Stable

TYPE UNSTABLE

```julia
x = ["a", 1]                    # variable with type 'Any'



function foo(x)
    y = 2 * x[2]

    [y * i for i in 1:100]
end
```

```
julia> @code_warntype foo(x)
```

```
MethodInstance for foo(::Vector{Any})
  from foo(x)
Arguments
  #self#::Core.Const(Main.foo)
  x::Vector{Any}
Locals
  #66::var"#66#67"
  y::Any
Body::Vector
1 ─ %1  = Main.:*::Core.Const(*)
    %2  = Base.getindex(x, 2)::Any
          (y = (%1)(2, %2))
    %4  = Main.:(var"#66#67")::Core.Const(var"#66#67")
    %5  = y::Any
    %6  = Core.typeof(%5)::DataType
    %7  = Core.apply_type(%4, %6)::Type{var"#66#67"{_A}} where _A
    %8  = y::Any
          (#66 = %new(%7, %8))
    %10 = #66::var"#66#67"
    %11 = (1:100)::Core.Const(1:100)
    %12 = Base.Generator(%10, %11)::Base.Generator{UnitRange{Int64}, F} where F<:var"#66#67"
    %13 = Base.collect(%12)::Vector
          return %13
```

The effectiveness of a barrier function ultimately hinges on how the function `foo` will be applied. In the given example, the barrier-function solution would be sufficient if `foo` is called only once. Instead, if `foo` is eventually called in a tight for-loop, the type instability of `2 * x[2]` would be incurred multiple times. In such cases, simultaneously addressing the type instability in `2 * x[2]` could entail substantial performance benefits.

FOOTNOTES

[1.] In this particular example, there's an easier solution, where `0` is substituted with `zero(x)`. The function `zero(x)` has been designed to return the additive identity (i.e., the null element) of `x`'s type.